OMPHALINA

Newsletter of FORAY NEWFOUNDLAND AND LABRADOR

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Foray Newfoundland and Labrador

is an amateur, volunteer-run, community, not-for-profit organization with a mission to organize enjoyable and informative amateur mushroom forays in Newfoundland and Labrador and disseminate the knowledge gained.

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We eagerly invite contributions to Omphalina, dealing with any aspect even remotely related to NL mushrooms. Authors are guaranteed instant fame—fortune to follow. _Issues are freely available to the public on the FNL website.

 No picture, no paper. Material should be original and should deal with the mycota of Newfoundland and Labrador and their concerns. Detailed Information for Authors is available on our website.

OMPHALINA

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CONTENT*

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Cover: A bit of a departure from the usual, this collage incorporates a portrait of Irene Mounce (Mycologia, with permission) and the stunning undersides of a specimen of her namesake, Fomitopsis mounceae (images: Maria Voitk). You'll forgive the Editor for placing them against a wall—no matter how beautifully painted that wall may be—as a symbolic nod to the challenges facing women in the early days of mycology. Image sources: Mycologia, with permission (Dr. Mounce); fungi (Maria Voitk); wall (bharath on Unsplash).

Message from the Editor

I admit I'm nowhere near competent with my Amanita IDs. If it's yellow, it's "*Amanita flavoconia*" to me. Not convinced? Check out Glynn's beautifully detailed sketches of this common NL mushroom in "The Bishop's Sketchbook", this issue.

Hello again, friend of fungi!

With voices near and far, from eager new learners to the old hats, I hope there is something for everyone in this issue. Our FNL President, Helen, has submitted a thoughtful review of the content in this issue on the next page, so I will let her tell you what to look forward to.

It was on Andrus Voitk's suggestion that we publish this issue to coincide with International Women's Day 2021, a suggestion for which I am grateful. We are so fortunate at Foray Newfoundland and Labador to have strong women leaders—from the Board, to our faculty experts, to the innovative and creative community members who share their expertise with us at our workshops and presentations—that we might forget that our mycological world was not, historically, always so welcoming. Perhaps this reminder offers perspective on how we can continue to make our commnities, our networks, our organizations more supportive and welcoming to everyone who shares our curiousity about the natural world.

Lastly, I'll end with an invitation to all of our readers: if you enjoy this publication and would like to contribute your skills to its continued (and perhaps more frequent) production, I invite you to please get in touch. We have more amazing content contributions than we can keep up with; my hard drive is in danger of becoming a private library of mushroom stories. Surely there is a mushroom enthusiast with future editorial aspirations out there somehwere....?

As always, Happy Hunting everyone!

Sara

Message from the President

Hello Foray NL members and friends,

Our contributors to this issue of Omphalina have, once again, done Foray NL proud. They bring you interesting reading to keep you entertained during this cold month of March, when we are hunkering down and dreaming of spring. This Omphalina is all the more welcome at this time because those of us in Newfoundland and Labrador, who have had a very lovely stretch of limited covid socializing, are into our second lockdown and in need of distractions.

I have had a happy morning previewing the articles, several of which have a bit of a poisonous thread stringing them together. Mark Feener's story is a good reminder for those of us with limited knowledge to recognize these limitations when offering help with mushroom IDs, and Jamie's white mushroom mysteries remind us that, for the curious, there is always more to learn. I think that if you are a *Leccinum* eater (or not) then Henry Mann et al.'s story and its accompanying references are worth a review before you eat your next haul. And as a bonus if, like me, you didn't know of the Bottom Brook Arboretum near Corner Brook, then you might decide to stop by for a visit, as I will try to next time I'm on my way to the ferry. As always Glynn Bishop's gorgeous paintings are an inspiration and maybe even better looking than the real thing. The offerings in the photo corner and the Notable Finds are delightful and will hopefully inspire more of you to send Sara your best fungal shots for a future issue.

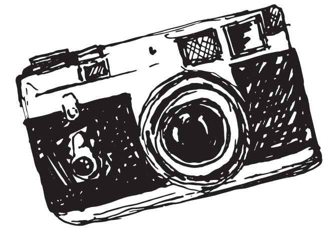
We are in luck, since March 8, the day of publishing this Omphalina, is also International Women's Day, thus we are celebrating with some articles highlighting significant historical contributions to the fields of mycology and microbiology made by women. Andrus Voitk, always a great story-teller, enchants us with the story of Irene Mounce who grew up around the 1900s in a remote mining village on Vancouver Island and later went on to make a brilliant investigation into *Fomitopsis*, which becomes a second story. These stories lead Andrus into an interesting account of Speciation and Species distribution. As if this isn't enough fodder for this issue, Gabby Riefesel regales us with yet another tale—this one about Fanny Hesse whose practical expertise lent us the perfect growth medium to culture microscopic organisms. Where would mycology be without it?

I hope you enjoy this issue. Thank-you to all of you—the contributors and their appreciative audience. Keep well everyone.

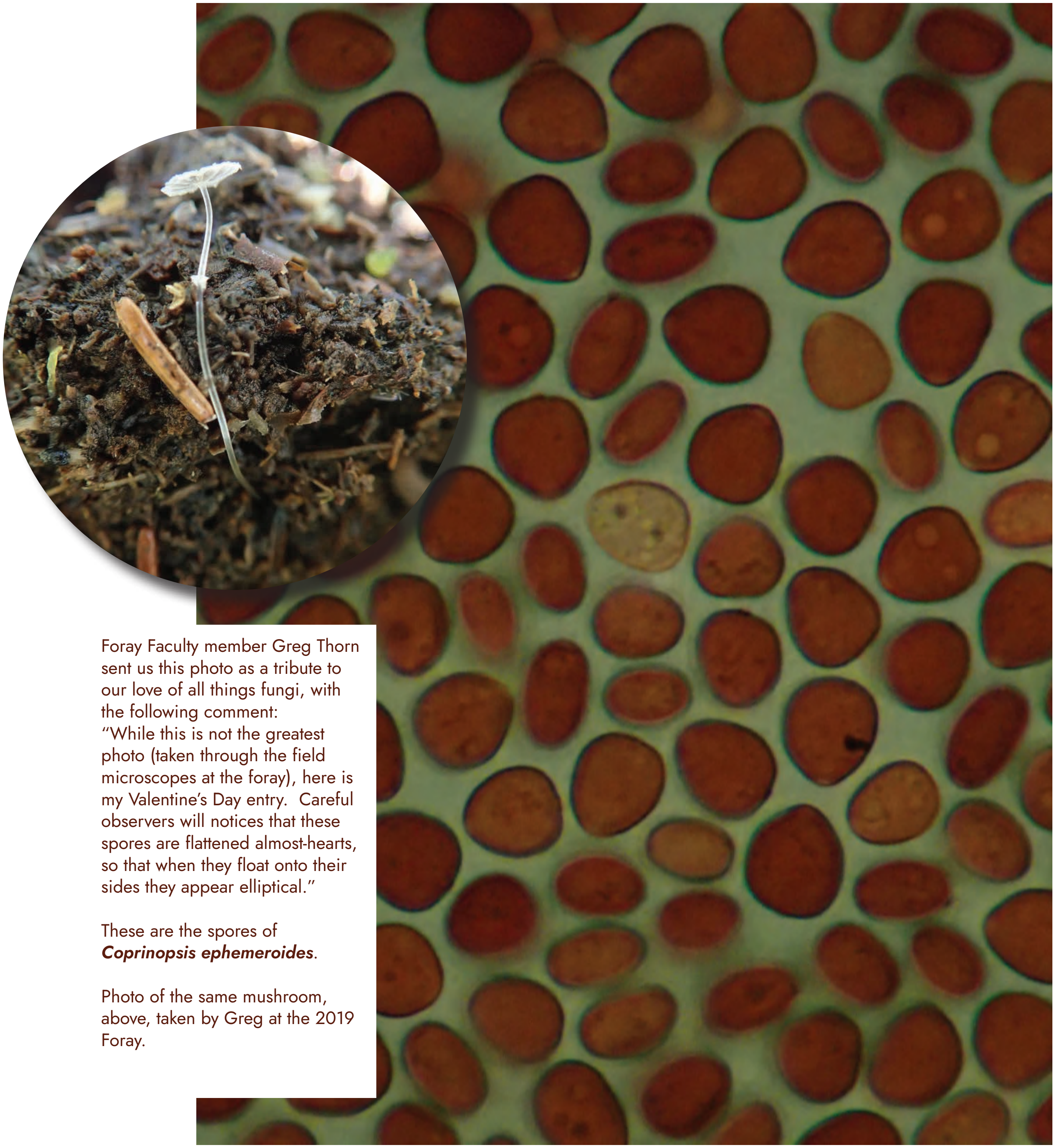
Helen Spencer

March 6, 2021

The Photographer's Corner



the weird & the wonderful from the wild



Foray Faculty member Greg Thorn sent us this photo as a tribute to our love of all things fungi, with the following comment:
"While this is not the greatest photo (taken through the field microscopes at the foray), here is my Valentine's Day entry. Careful observers will notices that these spores are flattened almost-hearts, so that when they float onto their sides they appear elliptical."

These are the spores of ***Coprinopsis ephemeroides***.

Photo of the same mushroom, above, taken by Greg at the 2019 Foray.

Irene Mounce and *Fomes (Fomitopsis) pinicola*

Maria's unique capture of this *F. mounceae* highlights the creamy white underside of a young specimen.

Andrus Voitk and Maria Voitk

This is a brief distillate of Irene Mounce's brilliant work with *Fomes* (*Fomitopsis*) *pinicola*. Conducting very meticulous and sophisticated studies that involved growing mycelia from single spores from various specimens, and experimenting with attempts to mate these mycelia in the laboratory, she discovered that the entity to which the epithet *pinicola* was applied across the Northern Hemisphere, is split into three groups: two in North America and one in Eurasia.[1] Both North American groups were sympatric, distributed coast to coast across the northern treeline, and were intersterile, but both would mate with the Eurasian group to some extent. At the time, she relied on the classical definition of species, which included inability to interbreed, and because all her three groups showed at least some degree of interbreeding with at least one other group, she was reluctant to declare them different species.

You may wonder why we consider this work stunning. As you will learn in the next two articles, the two North American groups are now considered different species, along with a third group found in the southwest of the continent that she did not survey.[2] No doubt you will immediately recognize this as a relatively common pattern of speciation in North America. For example, studies of the *matsutake-caligatum* complex,[3] and our own studies of *Polyozellus*,[4] showed very similar distribution patterns. So why is her work so special?

First of all, this continental distribution pattern was not at all known at that time. Nor were the principles governing evolution and speciation of fungi. Irene Mounce made her discoveries at a time when science barely understood what a cell nucleus was, well before the double helix, at a time when

DNA was much of a mystery. These days we extract and amplify DNA markers with primers using "automated" kits and analyze the results with very sophisticated (and "automated") computer software which produces trees, chooses the best or most likely, and performs statistical analyses to give us an idea of the significance of our findings. Instead of DNA, Irene Mounce germinated single spores from various collections, and studied the interbreeding of such mycelia in pure cultures—a major feat—and had the brilliance to interpret her results. An apt comparison might be to do very complicated mathematics longhand, without a slide rule, let alone a computer. Her results lay dormant for almost three-quarters of a century, waiting for reinterpretation in the light of changed concepts.

To me this has a personal aspect. I became aware of Mounce's work purely by accident. In 2006 we noticed some young conks on a birch tree not far from our house. We passed the tree regularly on our walks, and I watched the conks grow. They looked like *Fomitopsis pinicola*, but lacked the red band. I wasn't sure a band was always needed, as there seemed to be other similar pinicolas. After asking around for a few years, Greg Thorn suggested it might be a new species, *Fomitopsis ochracea*. I sent a conk to Leif Ryvarden, who confirmed the identification with DNA sequencing. When I looked it up, I learned the species was collected from aspen in Alberta in 2005, a year before I saw these in NL, and described in 2008 by Stockland and Ryvarden.[5] I reported the find in 2011,[6] and learned that this was the first record of it since its original description.

That led me to discover the work of Irene Mounce. I was enthralled, and tried to sell the idea of an update of this work, using current molecular techniques, to many taxonomists and polyporologists. Alas, I am a poor salesman, for I found no takers. Almost gave up.

One day in 2016 I read an article about this very subject by John-Erich Haight and collaborators, which showed the species in the *F. pinicola* complex just as Mounce had predicted.[7] Amazing! An exciting case of parallel evolution, coming up with the same idea all the way across the continent, in Alaska. I wrote John and sent him several of our specimens. He followed the first article up with a taxonomic work, describing the new species.[2] From there we learn that the name for the red-banded conk in our part of the world is not *F. pinicola*, but rather the new species, *F. mounceae*. Nice name, highly deserved. Use it henceforth!

In summary, we have two somewhat similar species of the *F. pinicola* complex, *F. mounceae* and *F. ochracea*, but

Two views of *Fomitopsis ochracea* (this page)

we do not have *F. pinicola* itself. I shall not describe our two species here because, even before we knew this and knew of the new name for what we had called *F. pinicola*, they have been featured several times on the pages of Omphalina. If you want to learn more about them, how they weep, how to tell them apart, check the Omphalina archives for the references listed here.[8–15]

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Two views of *Fomitopsis mounceae* (this page)

Irene Mounce

by Andrus Voitk

Figure 1: Irene Mounce *ca* 1935. Image copyright: Mycologia[4] reprinted with permission.

Irene Mounce (Fig. 1) was born in the Vancouver Island coal mining village of Cumberland on August 31, 1894. Her father oversaw the local lumber industry in the small community, letting her enjoy a privileged childhood. Mounce's schooling took her to the very first class of the brand new University of British Columbia (Fig. 2), in Vancouver, where she earned a BA in 1918, followed by an MA in 1920, accompanied by a Governor General's gold medal for extraordinary scholarship. Scholarships and her own earnings were to fund her remaining studies. From UBC she moved to work with Profs A. H. R. Buller and A. T. Cameron at the University of Manitoba, where she completed an MSc degree in 1922, and from there she moved to Toronto to work with Prof J. H. Faull. From Toronto she moved to the Division of Botany, Canadian Department of Agriculture in Ottawa, successfully defending a PhD thesis in 1929.

Behind this short and dry list of dates, places, mentors and degrees lies extraordinary achievement, all the more so, given her time and society, her village origins (even if somewhat better off than most), and, yes, definitely her gender. Irene Mounce, much as we all, was a product of her time and place. Despite other natural resources (coal, fur, gold, fishing), forestry was the main resource supporting the population of Vancouver Island. Long, straight spruce lumber was prized for aeroplane construction at the time because of its resilience and light weight. Thus, it was not surprising that the quality of this resource was a focus of commercial interest in British Columbia, and became the master's project for the curious scholar from Vancouver Island;[1] most

Figure 2: Irene Mounce (top row, left) at UBC in 1915. Irene's thesis was only the 8th completed at the newly-established university, making her a pioneer on a number of fronts.

Image source: University of British Columbia Archives, Photo collection, photographer unknown [UBC 1.1/1643]

disease of spruce was, of course, fungal, which brought this group of organisms to her attention for life.

Although the truly gifted make their own luck, even they profit from whatever smile fortune may bestow upon them. Irene Mounce worked hard to afford further study in a more mycologically attuned environment, but to end up in Winnipeg with Reginald Buller (Fig. 3) as her mentor, was great fortune that must have made a lasting contribution to her future work. In keeping with his somewhat romantic and iconoclastic image, Buller was a cultured, educated, gifted and erudite man, rounded well beyond mycology or natural history, and one of the more original thinkers of his time. During his studies Buller had worked for a year in Germany with Prof. Robert Hartig (Fig. 4), considered to be the father of forest pathology, that afforded Buller the opportunity to study mycelia and mating experiments.

Buller's concept of extramycological gender equality was even more ahead of his time. Co-founder of the Scientific Club of Winnipeg, he resigned membership when a visiting colleague was barred from attending because she was female. (The by-laws were changed quickly to avert scandal, permitting Buller to remain a member.) He took on disproportionately more female postgraduate students at a time when it was not "fashionable" for women to pursue academic careers, or even to encourage such pursuits. Irene Mounce was one of these women, earning one of the first MSc degrees awarded by the University of Manitoba. She did some of the early mating studies with mushrooms, not only the wood-rotting polypores, but also with species of *Coprinus*, laying a foundation for her future efforts. In Winnipeg today, Buller's legacy includes giving his name to the Botany Building, and, unwittingly, a slice of his brain to the paternalistic pathology collection (the rest of him was cremated, in accordance with his wishes for the entirety).

After the heady melding of minds and ideas in Winnipeg, Irene Mounce was ready for more independent investigation. She went to pursue further

Figure 3: Reginald Buller, poet and scientist.

Figure 4: Robert Hartig, first on the left (holding small boy), on an outing to Eberswalde. Bit of a boys' world for Ms Mounce.

Figure 5: Joseph Faull, 1937. Torontonian at Harvard. Image: Harvard University Herbaria/Botany Libraries.

studies under the tutelage of Joseph Faull (Fig. 5), Professor of Botany at the University of Toronto. It is not an accident that she moved to work with another tutor, who, like Buller, had also spent a year in Germany studying with Hartig; as a result Faull tutored some postgraduate students pursuing Hartig's methods of mycelial mating in the laboratory. In Toronto Faull is perhaps best known for editing a book on the natural history of the Toronto region,[2] to which he contributed not the chapter on mushrooms, but one of the earliest lists of lichens in the area, quoted to this day. A copy in good order can be found on the web for under $100 CAD. In 1928 Faull left Toronto to become Professor of Forest Pathology at Harvard and Irene Mounce joined the Division of Botany of the Canadian Department of Agriculture (Ottawa). There she had contact with other bright botanists and mycologists, and the opportunity to work with several keen young women pursuing similar interests. Out of these studies came her doctoral thesis, a brilliant investigation into the mating pattern of *Fomes* (now *Fomitopsis*) *pinicola*.[3]

How easy was it for a young girl from a remote mining village to reach the pinnacle of scientific achievement? Somebody as gifted as Irene Mounce gives a false impression: it seems too easy. But we do see glimpses of how difficult it really was, even for her. Not every academic mentor held attitudes like Buller. In a world where a visiting scientist is denied entry to a local scientific club meeting because of her gender, access to higher education for women may be unattainable, no matter the qualification. We need not look to primitive cultures for such difficulties. Look at Figure 4 again. If you did not know it depicted the dashing young Robert Hartig among a happy group of fellow academics (and an owl), you might be excused for thinking it might show happy Swiss voters after an election in any year up to 1971. That was the first year women were allowed to vote in Switzerland, i.e., after which women could also be expected on a photo of Swiss voters. Or perhaps you thought it showed a pre-2003 picture of the viola section of the Vienna philharmonic, relaxing on an outing to the countryside. This orchestra appointed its first woman member, a violist, in 2003—but only after economic pressure was applied by making government funding conditional on a move toward gender equality. For comparison, a photo of the Toronto Symphony Orchestra over 70 years earlier (Fig. 6) shows six women players (none of them in the viola section).

Not only was it difficult for a woman to pursue postgraduate scientific studies, a career in science held even more difficulties for women. Although Irene Mounce's work with *Fomitopsis* is the focus of our interest in this series of articles, as you can imagine, she went on to tackle several significant

City of Toronto Archives, Series 1569, s1569_fl0008_it0001

Figure 6: Toronto Symphony Orchestra 1931–1932, at Massey Hall. I can count at least six women. The year 1931 was a mere 55 years after Alexander Graham Bell patented the telephone, 28 years after the first flight by the Wright brothers, 2 years after Irene Mounce earned her PhD—and 22 years before Watson & Crick published the structure of DNA and 37 years before the term "double helix" was popularized by Watson—40 years before Swiss women could vote, 72 years before the first female player joined the Vienna Philharmonic and 90 years before this story is published.

projects during her employ by the Government of the Dominion of Canada. This all came to a quick end in 1945, when she married. Ginns states that she was "required to resign because of her marital status,[4] whereas Paterson in the Saanich News states, "employment of married women was forbidden in Canada until 1955."[5] I lived in Canada during that time, and do not recall such a law, because before 1955 my mother, very much married, was employed as a newspaper proof reader and later as a bank teller. Not at the same wage level as a man, of course, but at least seemingly openly and lawfully. Maybe my scheming parents hid their criminality successfully from my naïve eyes. Alternately, it is possible that given the number of returned soldiers needing employment after WWII, the employment policy of the federal government differed from industry, or perhaps just some more critical jobs, like mating mushrooms, were considered the domain of men or unmarried women of the Dominion.

Coincidence: this story was penned on International Women's Day, 2019.

Acknowledgements

I thank Jim Ginns, Tom Booth and Scott Redhead for helpful information. Although many sources were used, much of the factual information about Irene Mounce comes from the obituary written by Ginns[4].

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Speciation and species distribution

Andrus Voitk

photo: Timo Wielink on Unsplash

As mentioned in the preceding pages, a very common distribution pattern for North American species complexes is to have different species on the east and west coasts of the continent, in addition to some transcontinental species. To understand how such patterns develop, we need to understand speciation (evolution along diverging genetic lineages that results in new species). The first principle is that speciation cannot occur if there is interbreeding among the diverging lineages, because that would continue to mix all genetic material, constantly eliminating differences. Barriers to interbreeding can be either internal (i.e., erected by the organisms to prevent mixing of genetic material as they evolve), or external (e.g., geographic distance, ocean, barren areas, regions of inimical climate, etc.).

During a talk about at our 2012 foray, Nils Hallenberg told us that sympatric species (those that live in the same area; from *sym* = the same & *patria* = fatherland or home) cannot evolve along different genetic lineages without first erecting internal interbreeding barriers. Only with barriers to interbreeding in place, can separate lineages evolve side by side. Species with internal breeding barriers will be found to be intersterile on mating experiments in the laboratory. This is what Irene Mounce found about the two groups of *Fomitopsis* that extended across North America: they occupied the same space, and neither mated with the other. Such barriers had not occurred in Eurasia, and thus Irene Mounce found only a single group there.

Nils also explained that to erect barriers to each other, species must have some awareness of each other. Parapatric species (*para* = beside, apart from) grow in separate regions, separated by natural barriers that prevent genetic mixing, and can evolve in different directions without ever knowing the other exists. It would seem that genes controlling physiologic-ecologic characters evolve in response to environmental stimuli, but if the species are separated, genes controlling breeding have no stimulation to evolve. Again, this fit with Irene

Mounce's observation: both intersterile North American groups could interbreed with the Eurasian group in laboratory mating experiments. Because the North American groups were separated from the Eurasian one by the Atlantic Ocean—which acted as an effective barrier to their exchanging of genetic material or otherwise becoming aware of each other—they had had no stimulus to erect internal barriers to interbreeding.

Noting that the North American groups mated with the European ones, Irene Mounce quite correctly did not declare them separate species, because the definition of species at the time included inability to breed and produce fertile offspring. The theories expressed by Nils are the result of intervening investigations, which have shown that species definitions fitting mobile animals may not apply equally well to fixed fungi. Hence our current investigative tools enable us to state that the groups identified by Irene Mounce are currently recognized as bona fide distinct species.

Haight's studies identified a species of the *Fomitopsis pinicola* complex in southwestern North America, confined to the west of the continent. At the FNL 2006 foray, Ron Petersen described how it came about that the east and west coasts often host different species in a complex. During several cyclical periods of glaciation, most of Canada and northern USA was covered by an ice cap, up to thousands of meters thick, and was unable to support larger flora/mycota. Tundra extended beyond the ice cap, with the tree line much further. Sheltered ecological refugia, many far from the ice, hosted higher plants and their associated fungi. Two important refugia were the southern Appalachian Mountains and temperate Mexico. The ice receded during cyclical periods of global warming between the periods of glaciation, permitting flora and associated mycota to colonize newly ice-free habitats, marching slowly behind the retreating ice.

The Appalachian ecosystem followed the mountain chain northward, but the Mexican refugium seemed to bifurcate into western and eastern branches. Over the millennia the organisms moving along the east and west corridors encountered different environments, requiring new characteristics for survival. Successive warming and cooling and resultant movement re-mixed the new forms with those unchanged in the refugia. These southern "mixing pots" acted like species pumps, sending out new genetic combinations on each successive outward journey.

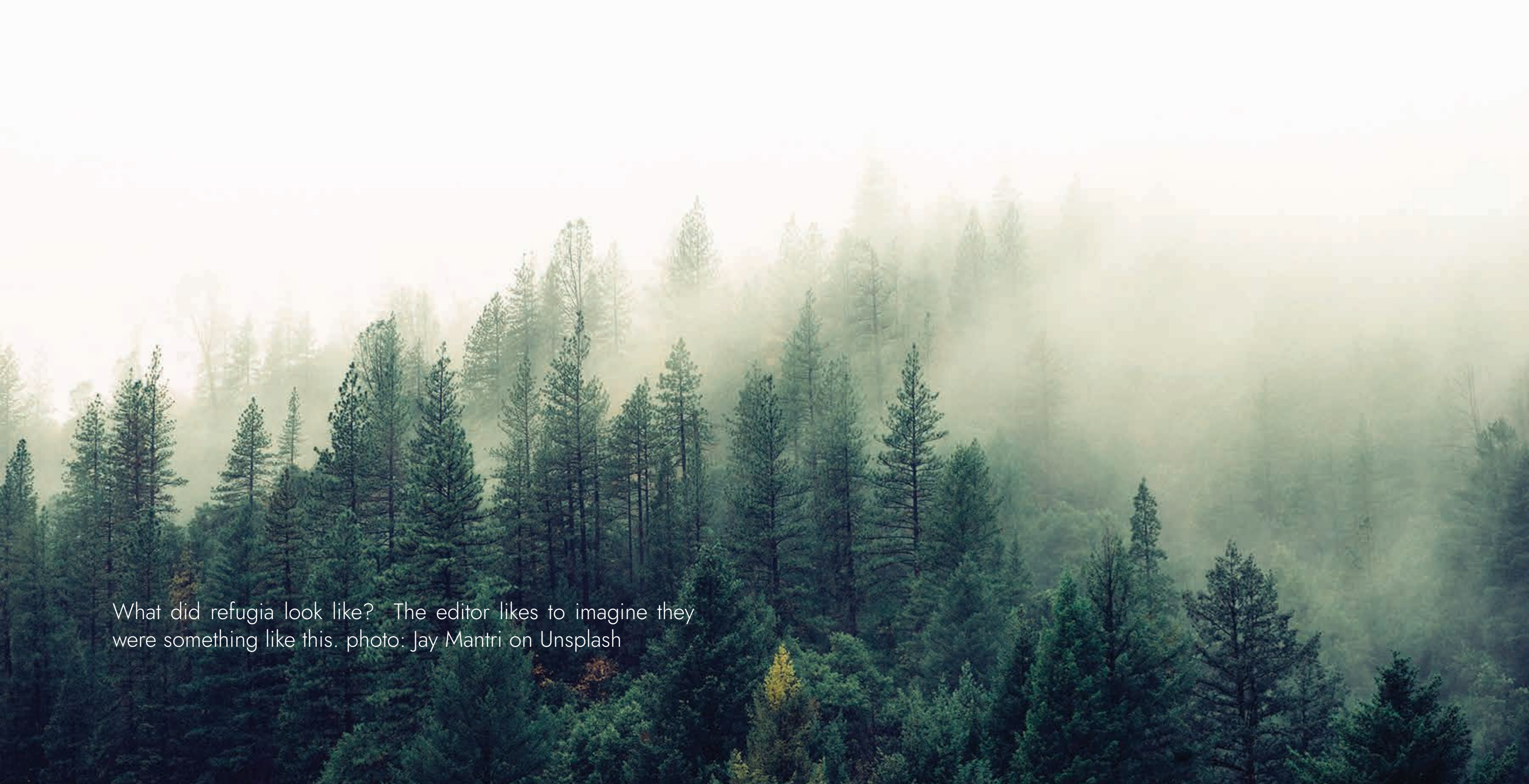
photo: Randall Meng on Unsplash

You may wonder why the species found in the southwest of the continent did not extend across the continent, like the two found by Mounce in the north. This was also covered during a talk about *Polyozellus* in 2017. Namely, fungi are not isolated organisms, living on their own, but are part of a system, linked to other organisms with whom they have interdependent relationships. Their action is either mycorrhizal, giving water and minerals to plants (most commonly trees) in exchange for sugars, or it is the decomposition of organic material, again with a variable choice of hosts. The spread of both kinds of fungi is limited by the spread of their hosts. The distribution of species of *Fomitopsis*, which depend on wood to decompose, is limited by their own climate/environment tolerance, the distribution of their host trees, and their host specificity.

The *Fomitopsis* species found in the southwest of the continent does not seem to thrive in northern climates, and is a decomposer of only conifers. Therefore, it is limited to conifers in a more temperate setting. Its extent to the east is limited by the eastward spread of coniferous woods, which ends at the treeless midwestern prairies and deserts. It has no way to reach eastern temperate conifer forests. The two species that Irene Mounce documented on the higher latitude eastern and western sides of the continent differ from the southwestern species in two important regards: i) they thrive in northern climates and ii) they use both coniferous and deciduous hosts. The northern treeline forms a contiguous band from east to west, connected by a narrow bridge north of the prairies. Therefore, any fungus able to tolerate this northern climate has a habitable pathway that extends from one shore to the other. Also, their use of both coniferous and deciduous hosts extends their ability to spread across the prairies, because despite the name, prairies are not exclusively treeless grassland: *Populus* woods occur in various prairie locations, and both species identified by Irene Mounce have been collected from *Populus*. A wider host choice facilitates their transcontinental spread.

As you see, our understanding of how fungal species evolve, and therefore the very concept of what defines a species, has evolved faster than these species themselves. And the amazing thing is that all this knowledge is not the secret purview of scientific cabals, but made freely available by leading scientists of our time to all members of FNL; and this regardless of gender, marital or amateur status.

So a word to the wise: keep your membership payments up!



What did refugia look like? The editor likes to imagine they were something like this. photo: Jay Mantri on Unsplash

From the Jam Jar to the Petri Dish:
FANNY HESSE AND THE DEVELOPMENT OF AGAR

image: Invisible Worlds exhibition — 2012 © M J Richardson (cc-by-sa/2.0)

Gabby Riefesel

The story of the development of agar, a now-ubiquitous growth medium for microbiological cultivation, is a good one to share for International Women's Day. This story reminds us that revolutionary contributions to science can come from anyone with a keen interest in the subject, so long as we're willing to listen!

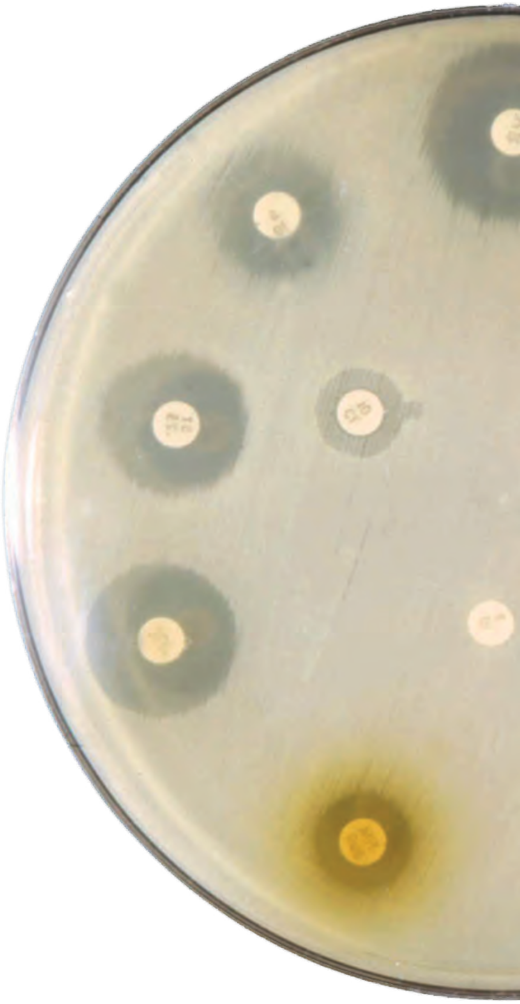
Angelina (Lina) Fanny Hesse worked as an unpaid technical assistant and scientific illustrator alongside her husband Dr. Walther Hesse. While assisting her husband as a medical technologist in a microbiology laboratory, Fanny solved one of the most significant challenges in the developing field of microbiology—how and where to grow organisms in the lab—with the seaweed extract known as agar.[1,2]

Fanny was a first-generation American, born in New York in 1850. Her father, Hinrich Gottfried Eilshemius, was a Dutch immigrant who ran a successful import business, and Fanny had a wealthy upbringing. When she was 15, Fanny's father sent her to finishing school in Switzerland to learn home economics and French. She was also able to travel abroad; while in Germany in 1872, she met her future husband, Dr. Walther Hesse. Walther was a physician and environmental toxicologist who studied airborne and waterborne illnesses. Once married, Fanny assisted with her husband's research, helping him prepare bacterial growth media and sterilizing lab equipment, while also raising and educating their three children. Fanny was an excellent artist, and she prepared illustrations of microscopic colonies and growth phases under high magnification that were used in Walther's publications.[1,2]

During Walther's sabbatical, he worked with Robert Koch, who is now considered one of the founders of modern microbiology, and credited with the discovery of the organisms responsible for tuberculosis and cholera, among others. In Koch's lab, the team aimed to achieve pure cultures derived from a single cell, but couldn't find a growth medium that was clear, solid, and wouldn't be metabolized by the growing microorganisms. Koch had tried gelatin and potato slices, but the impurity and environmental instability

Angelina (Lina) Fanny Hesse (1850–1934).

Walther Hesse (1846–1911). Images: public domain, Wikimedia Commons



of these media made microbial isolation challenging.[1,2,3] As you might expect, gelatin showed greater promise than potato slices, once processed and sterilized to produce a clear and solid substrate. But gelatin melted at 35º C, which rendered it effectively useless when studying bacterial growth at body temperature.

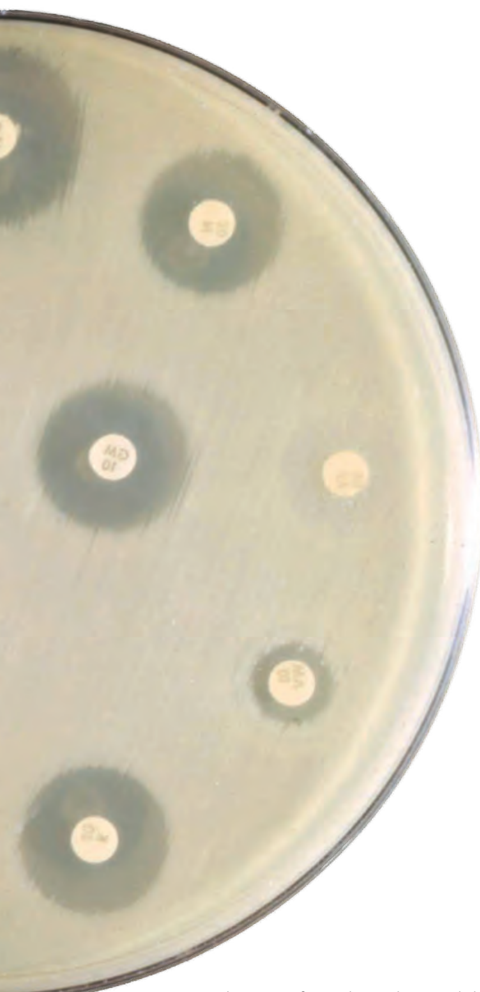
At the time, Fanny was using agar, a gelatinous seaweed extract, as a gelatin replacement in her kitchen. She had adopted the use of agar for making jams and jellies from an acquaintance who had spent time in Indonesia where agar (called "agar-agar", or "kanten" in Japanese) is used widely as a gelling agent in cuisine.[1,2] Fanny knew that agar was heat tolerant, and proposed it as a more stable growth medium.

Fanny's suggestion was a revolutionary moment for science; Robert Koch first isolated the *Mycobacterium tuberculosis* bacilli in 1882—on an agar plate— marking the first step towards developing successful treatment for the disease.[1,2,3] Unfortunately, Fanny's contribution was nearly lost to science, as Koch's early publications gave no credit to either Fanny or Walther Hesse for their invention of the agar medium. Fanny has only been widely recognized for this significant contribution to science in recent years.

The discovery of agar in the late 1600s is usually attributed to Minoya Tarozaemon, a Japanese innkeeper who is said to have left a red seaweed jelly dish for several cold nights, and noticed a clear jelly forming on the surface. Curious about this new, clear jelly with less seaweed odor, he brought it back to his kitchen. The rest is history!

Drying Tengusa seaweed, *Gelidium corneum*, at Izu, Japan for the commercial market. Tengusa is one of the primary sources of agar, destined for either the kitchen or the labraratory. Photo: Anonymous via Wikimedia Commons (CC BY-SA 3.0)

Today, agar gel is used in laboratories around the world as the standard culture medium in microbiology. Agar is a polysaccharide mixture derived from several species of red algae. Agar is resistant to microbial digestion, thermally stable, and can be sterilized and stored for long periods of time. A key characteristic of agar is thermal hysteresis; once heated and cooled below 45º C, agar can be heated up to 90º C without melting. Agar isn't only suitable for bacterial cultures. It is a choice growth medium for fungal cultures as well, leading to the development of medicines derived from fungi, including mitotic inhibitors used to fight cancer, antibiotics derived from penicillin, statins used for lowering cholesterol, vitamin development, and many others.[2,4,5,6]



Inoculation of *Escherichia coli* by *Penicillium sp.* Image: USCDCP on Pixnio

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Note the ragged cap cuticle extending beyond the cap edge; one of this Leccinum's distinguishing features. Black stem scabers are very prominent.

A Bottom Brook Leccinum

Henry Mann, Phyllis Mann, Maria Voitk, and Andrus Voitk

The Bottom Brook Arboretum is a 12 hectare "tree garden" located 60 km roughly south of Corner Brook on the Trans Canada Highway (TCH) established in 1967 by the Canadian Forest Service[1]. Exotic and native tree species from across North America, Europe and Asia were planted in blocks to determine their adaptability to the growing conditions in this western forest region. Tree species blocks are all labelled and a walking path loops throughout the stands, providing for a pleasant ramble within this patchwork forest (Fig. 1). Not only does the site provide foresters with a unique outdoor laboratory, it is also a mushroom hunter's dream.

Many mushrooms form mycorrhizal associations with tree roots. Some are very specific about the species of trees they will grow with, while others are generalists, forming associations with a wider group of species. A careful observer can recognize such associations while passing through the tree stands of this forest microcosm. The genus *Leccinum* is one of the bolete group of pored mushrooms which tends to be particularly choosy about the tree species with which its species will hobnob.

On a pleasant September 22, 2020 autumn day our group meandered along the main pathway near the entrance, having already encountered some interesting species. As we neared the European Birch stand a mass of large brownish-capped mushrooms came into view. Someone shouted "KING BOLETES!" as we rushed into the birches. Some of the mushrooms appeared to "march" in lines following the birch roots (Fig. 2).

Close examination revealed these were not *Boletus edulis*, but one of the leccinums, as evidenced by the prominent black scabers on the stems (Fig. 3). Caps were large, up to 20 cm across, brownish with a slight orange tinge, the surface cuticle extending slightly beyond the cap margin as ragged flaps (Fig. 4; header image). The pore surface was pale grey in younger individuals to olive-beige in older specimens. When sliced through, the flesh was white, but very slowly began staining to grey or black (Fig. 5). The pore surface also stained somewhat upon bruising.

This birch associate *Leccinum* did not appear in any of the other arboretum stands, even the other birch stands—and not even another European Birch stand—which brings other questions to mind that need not be discussed here. Such a large prominent and prolific tree-specific mushroom should be easy to identify, at least that seemed to be a reasonable assumption.

However, as Nature often reminds us, this task is not always so simple. A recent article[2] described the difficulties with *Leccinum*s with the following statement, "*.....attempts to quantify the North American Red- and Orange-capped Leccinums can only be described as a wet, hot mess.*" What we have is a large group of described *Leccinum* species, some of European and some of North American vintage, whose features are so similar and overlapping that uncertainty exists as to whether they are all or partly synonyms. Minor features such as cap colour used to delimit the named taxa are questionably valid. To separate the entities into "good" species, future DNA studies will certainly be needed to shed some light on these matters.

Of the many described species such as *L. atrostipitatum, aurantiacum*, *versipelle, testaceoscaber, boreale*, etc., we decided our Bottom Brook Arboretum species is *L. atrostipitatum*, the Dark-stalked Bolete, which is probably synonymous with at least *L. versipelle* and *L. testaceoscaber*. Descriptions in the literature seem to fit our specimen very well.

Many sources suggest that all the leccinums are good edibles[3], however, a few reports exist of toxicity associated with the group[4,5]. Serious toxicity has been reported with their ingestion, mostly from areas west of the Great Plains. However, occasional reports have also come from east of the prairies,

Figure 1: (A) Bottom Brook Arboretum entrance; (B) European Birch block sign; (C) Block plan for the Arboretum.

Figure 2: Mushrooms following a birch root route

Figure 3: *Leccinum atrostipitatum*. Cap colours do not always photograph precisely compared to visual viewing which appeared slightly more orange-brown than this photo. Note a European White Birch (*Betula pendula/verrucosa*) leaf on a mushroom cap.

even with an associated mortality, although it is not clear whether leccinums were responsible or mortality was causally related to their ingestion. To our knowledge, we do not have toxic species of *Leccinum* in the province, and some of us have enjoyed red-capped leccinums here, including a small amount of these under discussion, with no harmful effect. Regardless, individuals should never trust general statements of mushroom edibility in the literature as a guide to their personal reactivity and safety.

Perhaps we might have enjoyed more of the Bottom Brook Leccinums, however, the amazing season of *Boletus edulis* here on the west coast in 2020 year had satiated our need for more mushroom cuisine. But, perhaps next year?

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Figure 4: A specimen with slightly more orange colouration, even some bluish.

Figure 5: Upon cutting, the fresh white flesh very slowly will begin to stain grey to black

White(ish) Mushroom with Free Gills and a Ring? Not always an *Amanita*...

Jamie Graham

During the off-season (summer) when there is no snow around in Corner Brook, Francine and I walk most days. We often walk a two kilometer loop from our home on Raymond around West Valley and MacPherson. When we have more time we do a 3 or 4 km loop involving Elswick, Cobb Lane and the paths by Corner Brook Stream. We usually look at trees and sunsets, but in mushroom season we tend to look down rather than up. The first week of September we found several interesting lawn mushroom specimens.

Mystery mushroom #1 with a white cap, dark gills, and brown spore print: *Agaricus sp.*

The first was a group of mushrooms on Elswick. They had tan caps, about 7 cm in diameter and dark gills, later shown to have a dark brown spore print. The striking features were free gills and a ring which we know to be hallmarks of *Amanita* species. Obviously, because of the dark spores, these were not *Amanita*.

Two days later we found a white mushroom on Raymond. It measured 6 cm across, with a smooth tan cap and white gills (and white spore print). It also had free gills and a ring. I wondered whether this was an *Amanita* but did not see a cup at the base.

Mystery mushroom #2 with a white cap, white gills, and a white spore print: *Leucoagaricus leucothites*

The most unusual mushroom turned up later, also on Raymond. From a distance I thought it was a golf ball. It looked like a pretty parasol, about 5 cm across, with a white to grey slightly cracked cap. It also had free white gills and the ring was still partly attached to the cap. I could not find a cup in the grass and later I was unsuccessful in getting a spore print.

I did not think any of these mushrooms were a species of *Amanita*. I sent the photos to Andrus Voitk. He identified the Elswick mushroom as a species of *Agaricus* (possibly *A. micromegethus*) and the first Raymond mushroom as *Leucoagaricus leucothites*.

The Raymond parasol mushroom does not look like a *Lepiota*, is not *Chlorophyllum* (former *Macrolepiota*) *rhacodes*, and we have no *Macrolepiota procera*, so this mushroom remained a mystery for him. The gills have a slightly pinkish tinge, which is often associated with *Amanitas*, but the stem is not of an *Amanita* either. The national team was mustered.

Renée Lebeuf said she could not think of anything else than *Leucoagaricus leucothites*, its cap possibly cracked from exposure to the sun. She knew of two other white lepiotas (*Lepiota erminea*, *Leucocoprinus cepistipes*), but they look different. Greg Thorn said *Leucoagaricus leucothites* was his guess, too. He has seen specimens with grey-scurfy-scaly caps like these on several occasions. The stem and ring fit well, as does the colour of the gills.

So there you have it: **three different mushrooms (two species) with free gills and a ring, and not one *Amanita!***

Spore prints are excellent pieces of evidence in any riveting mushroom mystery. A good detective would take note of the dark spores from the *Agaricus* (left) and the white print of the *Leucoagaricus leucothites* (right).

Extra credit-worthy mystery mushroom #3 with a parasol-shaped, scurfy-scaly white cap, and white gills: a sunburned *Leucoagaricus leucothites?*

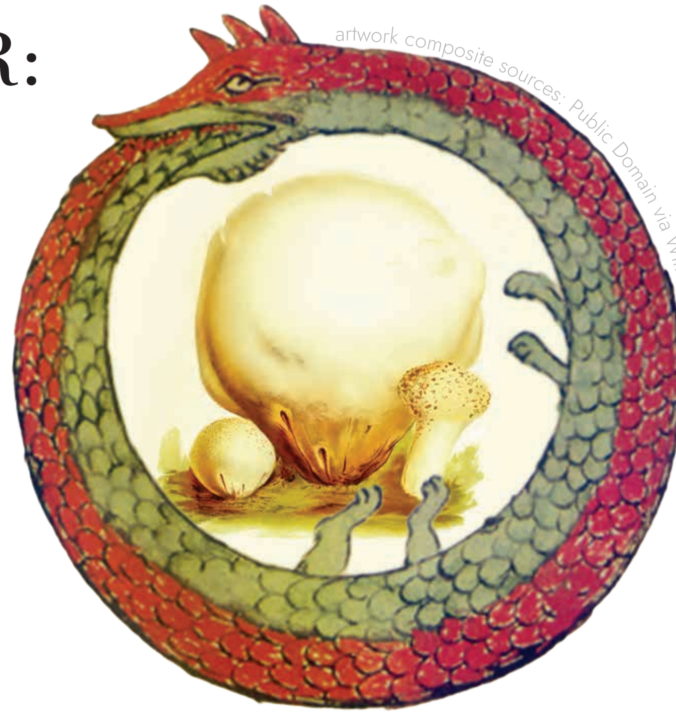
Editor's Note:

Your former AND current Editors would like to both remind you, dear reader, that although Jamie is fortunate enough to have every mushroom <u>except</u> *Amanita* in his neighborhood, that may not be the case in yours. Stay smart around those white caps!

Reflections from the ER:

A little knowledge can go a long way, but we're smartest together.

by Mark Feener



artwork composite sources: Public Domain via Wikimedia Commons

I got interested in mushrooms as a hobby. I find fungi very fascinating—the way they work, how much we don't know, the variety. Fungi are endlessly interesting to learn about, even for someone who has not had any formal mycological education.   You can spend hours reading, forget half of it and weeks later stumble across something in the woods that reminds you of something you've read, and BANG, you're Googling again! Like a weird ouroboros, the more you know, the more you realize you're only an inch or two farther from ignorance.

Basic mushroom learning can mature along a number of paths: 1) general interest and becoming a hobby forager; 2) becoming a professional forager; 3) becoming a professional mycologist; or 4) being considered "that guy" in your circle of friends and sharing your mushroom knowledge with anyone who will listen. *Actually, path 4 applies to all of us, doesn't it?* I'm a lucky guy along path 1 who can take his time learning about mushrooms because I have a job and I don't rely on mushrooms to put food on the table. I have a small hobby business with my best friend Toby. We forage for mushrooms and plants and make a little money doing something we find extremely rewarding and interesting.

Helping with the identification for a potential mushroom poisoning would not have been something I would have ever felt qualified for. I would consider that the realm of the professional, or at least a more experienced identifier. But then on a Sunday in early July of last year, a nurse from the hospital contacted my wife and asked her if I could help with exactly that: a young child in our community had taken a bite of a mushroom at a local park and now the family was at the emergency department.

Quick spoiler: *everyone was fine*. Turns out, the mushroom in question was an edible puffball and there was no danger except for the obvious emotions running high in the worried parents!

When you hear that a kid just ate a mushroom at a park and the parents fear for the worst, you tend to move faster than would be entirely wise. My story here today is just a warning about letting the high pressure of a suspected poisoning trap you into going beyond your reach. I hope it serves as a reminder that our shared community and knowledge network is valuable and stronger than any single person.

Here's what happened:

After looking at some pictures that were sent to my phone (Fig. 1), I grabbed every mushroom book I own and may have broken some traffic laws getting to the hospital. Even though my first thought was "it's a puffball", the small white "buttons" of poisonous *Amanita* like the death cap or the destroying angel can bear a striking resemblance to puffballs (Fig 2).   At the hospital, I met the nurse in the lobby and had a look at what remained of the mushroom in question. The family had been told, variously, that "it might be nothing", "it could be fatal", "symptoms could take 36 hours to two weeks to appear". They were understandably frantic for clear information. And it was obviously a puffball. The inside was entirely white and homogeneous, with no differentiation into developing cap/gills and stem that would suggest a button *Amanita*. The outside skin was rubbery and around a millimeter thick, sporting a slightly tannish color.

I told the nurse that it was a safe and edible mushroom and there was nothing to worry about. I felt thankful that I could relieve the family's fear and uncertainty; best outcome possible behind leaving it alone in the first place.

<u>In hindsight, here's what I wish had happened:</u>

Call in my mushroom cavalry. Call everyone I have contact information for who is involved with mushrooms and get any information we know into the hands of folks I consider experts. I have knowledgeable mushrooms contacts who identify mushrooms for a living, and some who have been at it since I was flying jumps on my pedal bike. But in the heat of the moment, I just panicked and did the first thing that came to my mind—rush in and try to help. I'd like to think that if the mushroom identity had been less clear, I would have started shaking trees. But reflecting now, I think the best course of action would have been to call on my mushroom network first... and then run in with their collective insight in my back pocket.

Everyone got lucky in this story. I hope the family won't have long-term mycophobia... but I'd understand if they did. I get to feel good about identifying a puffball, more so than usual! And lastly, I hope I can pass along a word of advice if you find yourself in a similar situation: we're smarter together. If you think this mushroom stuff is complicated now, just imagine if everyone kept their research secret, like some kind of mushroom Fight Club.

Editor's Note: Mark's story is a good reminder that being "that guy" among your friends comes with both risks and rewards. To learn more about poisonous mushrooms in Newfoundland and Labrador, check out our back issues of Omphalina, and become a member of Foray Newfoundland and Labrador. Newsletters and circulated warnings, such as this, about potentially life-threatening poisonings (and thankfully safe stories) will help you stay knowledgeable about potentially dangerous mushrooms. Andrus Voitk's *A Little Illustrated Book of Common Mushrooms of Newfoundland and Labrador* (2007; The Gros Morne Cooperating Association) is also a handy, pocket-sized reference.

And please don't underestimate how much more effective learning about mushroom identification is in person—when you can observe mushrooms up close and personal with faculty and experts—than from apps or social media sites. We hope you'll consider joining us at our fall Foray sometime soon so we can all keep building our mushroom knowledge network together.

Figure 1: The offending mushroom in question at the hospital.

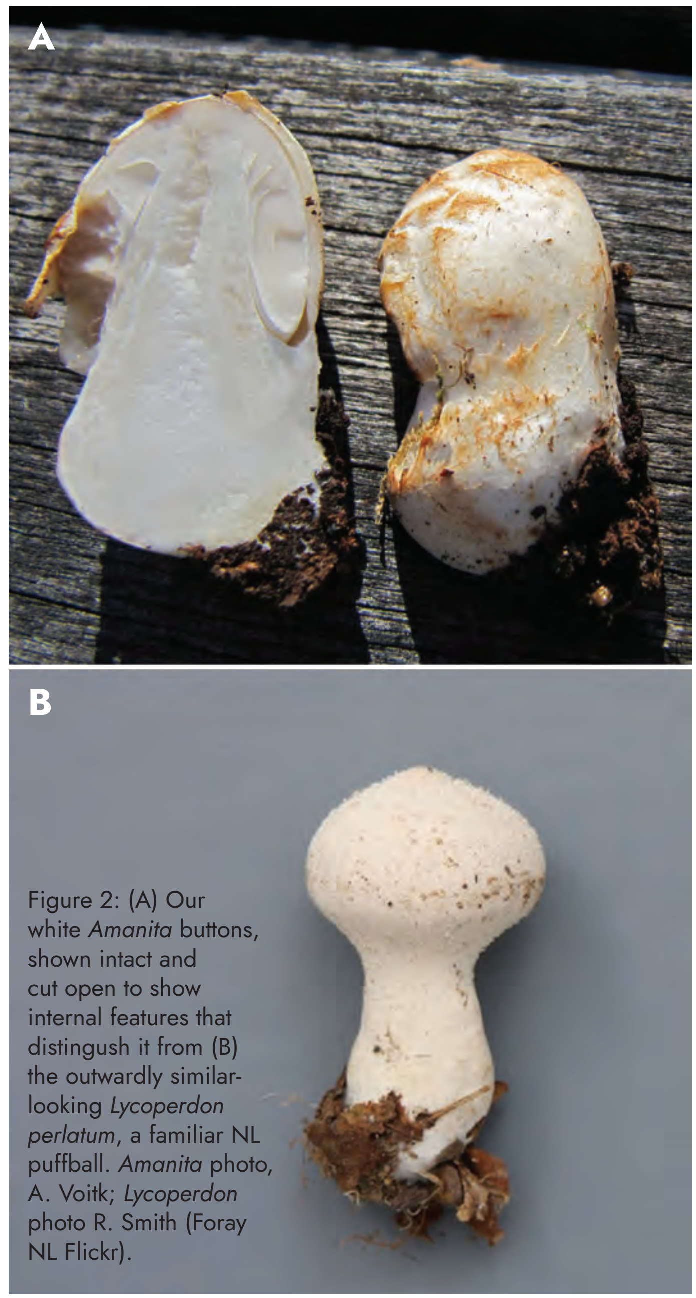


Figure 2: (A) Our white *Amanita* buttons, shown intact and cut open to show internal features that distingush it from (B) the outwardly similar-looking *Lycoperdon perlatum*, a familiar NL puffball. *Amanita* photo, A. Voitk; *Lycoperdon* photo R. Smith (Foray NL Flickr).

NOTABLE FINDS

Thick hair ice on a downed branch in SE Labrador. photo: Russell Pilgrim

Thanks go to Russell Pilgrim and Isabella St. John for sharing these images of a surreal feature you might have seen this winter. Called “hair ice”, or “silk frost”, this remarkable phenomenon is made possible by living fungi in the twigs or branches on which the ice forms.

Researchers in Switzerland[1] recently explored the potential role of fungi in hair ice formation, a theory first proposed nearly 100 years earlier.[2] They found a range of both Asco- and Basidiomycota in or on wood that grew hair ice in winter, but suggest that ***Exidiopsis effusa***, ubiquitous in all of their wood samples, may be the fungal culprit.

Hair ice forms on calm, cold, humid winter nights. If the air is too dry or windy, the ice will sublimate and no “hairs” grow. It forms on bark-free patches or gaps on wood twigs. The precise mechanism of ice growth is not entirely clear, but it appears that heat and gases produced by metabolic activity of the fungus pushes water outward form the wood, where it freezes just at the wood ray—air interface, where the “hairs” grow. The fungus may also play the part of a mild antifreeze agent that helps to maintain the ice’s hirsute character by inhibiting ice recrystallization into larger, coarser shapes. Learn more by reading Jan Thornhill’s account of experiments efforts to grow hair ice in Mycologia[3].

— Sara Jenkins

You can clearly see the twig through this icy mane. photo: Isabella St. John

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The Bishop's Sketchbook

Art by Glynn Bishop

A rare portrait of the artist at work!

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